

Information Management

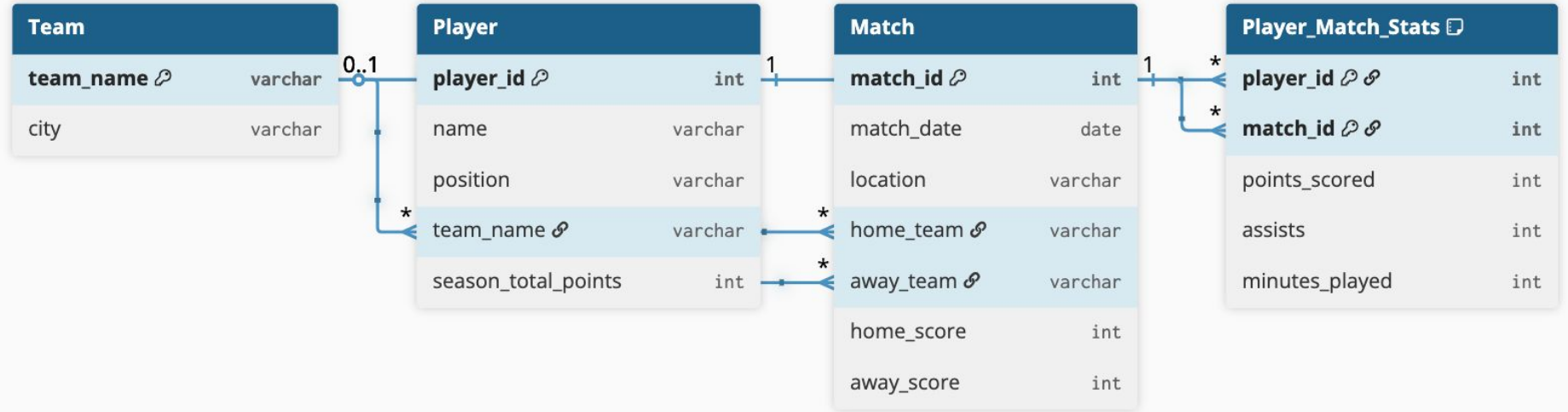
Assignment 4

MAITREYI PILLAI

Q1. Explain the difference between a weak and a strong entity set. Use an example other than the one in Chapter 6 to illustrate. (Consult Ch. 6, 6.5.3)

- Strong Entity Set: An entity set that possesses a primary key and can exist independently of other entities.
 - Weak Entity Set: An entity set that does not have a primary key of its own and depends on a strong identifying entity set for its existence.
 - Example: In a Hotel Management System for example suppose a Hotel is a strong entity (identified by Hotel_ID). Room is a weak entity because room numbers (e.g., "Room 101") are not unique globally and only have meaning when associated with a specific Hotel_ID.
-

Q2



Q3.

3a. Natural Join Explanation

- The original query joins takes and student to find average credits for sections with at least two students in 2017.
- A natural join with the section table matches rows on common attributes (course_id, sec_id, semester, year).
- Because every course_id and sec_id present in the takes table must already exist in the section table to satisfy referential integrity, joining them does not filter out any additional records

3b. `sql SELECT student.ID FROM student LEFT OUTER JOIN takes ON student.ID = takes.ID
WHERE takes.course_id IS NULL`

3c. `SELECT employee.ID FROM employee NATURAL LEFT OUTER JOIN manages WHERE
manager_id IS NULL;`
